



6.4.2 Numerical stability of LU factorization

6.4.2 Backward error of LU factorization

fit width



The numerical stability of various LU factorization algorithms as well as the triangular solve algorithms can be found in standard graduate level numerical linear algebra texts [19] [22]. Of particular interest may be the analysis of the Crout variant of LU factorization 5.5.1.4 in

- [6] Paolo Bientinesi, Robert A. van de Geijn, Goal-Oriented and Modular Stability Analysis, SIAM Journal on Matrix Analysis and Applications , Volume 32 Issue 1, February 2011.
- [7] Paolo Bientinesi, Robert A. van de Geijn, The Science of Deriving Stability Analyses, FLAME Working Note #33. Aachen Institute for Computational Engineering Sciences, RWTH Aachen. TR AICES-2008-2. November 2008. (Technical report version with exercises.)

since these papers use the same notation as we use in our notes. Here is the pertinent result from those papers:

Theorem 6.4.2.1. Backward error of Crout variant for LU factorization. *Let $A \in \mathbb{R}^{n \times n}$ and let the LU factorization of A be computed via the Crout variant, yielding approximate factors $\tilde{L}$ and $\tilde{U}$. Then*

$$(A + \Delta A) = \tilde{L}\tilde{U} \quad \text{with} \quad |\Delta A| \leq \gamma_n |\tilde{L}| |\tilde{U}|.$$